Supplementary Material: Catalan-Exponential Priors for Bayesian Decision Trees: WAIC Consistency, Posterior Concentration, and PAC-Bayes Guarantees

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This document contains three supplementary appendices:

- Appendix A: Detailed balance of the RJMCMC Birth/Death moves.
- Appendix B: ECE calibration bounds for DM-leaf BDTs.
- Appendix C: Decision-theoretic analysis under asymmetric loss.

A RJMCMC Detailed Balance

A.1 Birth and Death moves

The JBDT sampler of Schetinin and Jakaite (2025) uses four RJMCMC (Green, 1995) moves; all threshold proposals use the *local* data range at the node being modified (main paper Section 2.4, Eq. (7)).

Birth (p_B): select a uniformly random leaf ℓ, draw a split variable $v \sim \text{Unif}\{1, \dots, m\}$, and draw a threshold $q \sim \text{Unif}(a, b)$ where $[a, b] = [\min_{i \in \ell} x_{iv}, \max_{i \in \ell} x_{iv}]$ is the range of variable v over the N_ℓ training samples that have descended to leaf ℓ. The proposal density is $g_j(q) = 1/(b - a)$. *Sweeping for Birth*: if either child would contain fewer than $p_{\min}$ training points the proposal is immediately *rejected*; no death-move collapse is performed.

Death (p_D): select a uniformly random prunable pair (sibling leaves sharing a prunable parent, of which there are r_T), and propose merging them.

Change-Split (p_{CS}): select a uniformly random internal node η, draw a new split variable $v' \sim \text{Unif}\{1, \dots, m\}$, and draw $q' \sim \text{Unif}(a', b')$ from the local range of v' over all N_η training samples in η's subtree.

Change-Rule (p_{CR}): select a uniformly random internal node η, and draw $q' \sim \mathcal{N}'(q, \sigma^2, [a, b])$, a Gaussian truncated to the local range $[a, b]$ of the existing variable v over η's subtree.

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Sweeping for Change moves (Schetinin and Jakaite, 2025): after applying the proposed (v', q') to node η, count underpopulated leaves $n_0 = \#\{j \in L(T') : N_j < p_{\min}\}$:

- $n_0 = 0$: proceed to MH evaluation;
- $n_0 = 1$: *collapse* the unique underpopulated sibling pair to a single leaf (Death move), yielding a valid $k-1$-leaf tree;
- $n_0 > 1$: *reject* (parents differ; collapse would break reversibility).

Remark A.1. Our Python implementation simplifies the $n_0 = 1$ branch by always *rejecting* ($n_0 \geq 1$), which preserves detailed balance but accepts fewer Change proposals than Algorithm 2 of Schetinin and Jakaite (2025).

A.2 Detailed balance for Birth/Death

Theorem A.2 (Detailed balance)**.** *Under the Catalan-exponential prior* (6)*, the Birth and Death proposals satisfy the detailed balance condition*

$$\log \alpha_B(T \to T') + \log \alpha_D(T' \to T) = 0 \tag{1}$$

for any tree T with k leaves and its Birth-proposal T' with $k+1$ leaves.

Proof. The log-acceptance ratios decompose as

$$\begin{aligned}
\log \alpha_B &= \underbrace{\Delta \log p(\mathcal{D} \mid T')}_{\text{likelihood}} + \underbrace{\log \pi(T')/\pi(T)}_{\text{prior}} + \underbrace{\log q_D(T' \to T)/q_B(T \to T')}_{\text{proposal}}, \\
\log \alpha_D &= \underbrace{-\Delta \log p(\mathcal{D} \mid T')}_{\text{likelihood}} + \underbrace{\log \pi(T)/\pi(T')}_{\text{prior}} + \underbrace{\log q_B(T \to T')/q_D(T' \to T)}_{\text{proposal}}.
\end{aligned}$$

Clearly the likelihood terms cancel. For the prior terms, from Section 2.3 of the main paper:

$$\begin{aligned}
\log_{\text{prior_birth}}(k, \gamma) &= \log \pi(k+1)/\pi(k) = -\gamma - \log\left[\tfrac{2(2k-1)}{k+1}\right], \\
\log_{\text{prior_death}}(k+1, \gamma) &= \log \pi(k)/\pi(k+1) = +\gamma + \log\left[\tfrac{2(2k-1)}{k+1}\right],
\end{aligned}$$

so the prior terms sum to zero. For the proposal terms, let $[a, b]$ denote the local range of the split variable at the chosen leaf (main paper Eq. (7)). The Birth proposal probability is $q_B(T \to T') = p_B \cdot (1/k) \cdot (1/m) \cdot 1/(b-a)$, where $1/(b-a)$ is the density $g_j(q)$ of the uniform threshold draw over the local range. The Death proposal probability is $q_D(T' \to T) = p_D \cdot (1/r_{T'})$, where $r_{T'}$ is the number of prunable pairs in T'. Correspondingly, $q_D(T' \to T)/q_B(T \to T') = (p_D/p_B) \cdot k \cdot m \cdot (b-a)/r_{T'}$, and $q_B(T \to T')/q_D(T' \to T) = (p_B/p_D) \cdot r_{T'}/(k \cdot m \cdot (b-a))$, giving $\log q_D/q_B + \log q_B/q_D = 0$. $\square$

Closed-form birth-prior contribution. Using $C_{k-1}/C_k = (k+1)/[2(2k-1)]$ (Proposition 4.1b):

$$\log_{\text{prior\_birth}}(k, \gamma) = -\gamma - \log\left[\frac{2(2k-1)}{k+1}\right]. \tag{2}$$

This is always negative (monotone decreasing from $-\gamma$ at $k = 1$ to $-\gamma - \log 4$ as $k \to \infty$), so the Catalan prior always penalises Birth, with an additional structure-dependent penalty $-\log[2(2k-1)/(k+1)]$ beyond the exponential term $-\gamma$. The total Catalan complexity regularisation range is $\log 4 \approx 1.386$ nats.

A.3 Chain convergence

Numerical verification: with $N = 80$ observations, $k^* = 3$ true leaves, $C = 2$, $\gamma = 1$, the pure-k RJMCMC chain over 50,000 steps achieves $\text{TVD}(\hat{\Pi}_N, \Pi_N) = 4 \times 10^{-5}$, decaying as $O(1/\sqrt{S})$ with number of steps S, as expected for MCMC CLT.

B ECE Calibration Bounds

B.1 Per-leaf MSE bound

Proposition B.3 (DM calibration MSE)**.** *For leaf j with N_j i.i.d. observations from p_j, Dirichlet prior* $\text{Dir}(\alpha, \ldots, \alpha)$ *($\alpha_0 = C\alpha$):*

$$\mathbb{E}\left[\|\hat{p}_j - p_j\|^2\right] = \sum_c \frac{N_j p_{jc}(1 - p_{jc}) + \alpha_0^2(1/C - p_{jc})^2}{(N_j + \alpha_0)^2}. \tag{3}$$

For large N_j: $\mathbb{E}[\|\hat{p}_j - p_j\|^2] \approx \sigma^2_{\max}/N_j$, where $\sigma^2_{\max} = \max_{j,c} p_{jc}(1 - p_{jc}) \le 1/4$.

Proof. Direct computation from the Dirichlet posterior mean and variance. $\mathbb{E}[\hat{p}_{jc}] = (N_j p_{jc} + \alpha)/(N_j + \alpha_0)$, $\text{Var}[\hat{p}_{jc}] = N_j p_{jc}(1 - p_{jc})/(N_j + \alpha_0)^2 + O(N_j^{-2})$. Combining bias^2 and variance gives (3). □

B.2 Prior-averaged ECE bound

Proposition B.4 (Prior-averaged ECE bound)**.** *For a k-leaf BDT with balanced leaves ($N_j = N/k$):*

$$\text{ECE}(k) \le \sqrt{\frac{C\,\sigma^2_{\max}\,k}{N}}. \tag{4}$$

Taking expectation over $k \sim \pi$:

$$\mathbb{E}_k[\text{ECE}] \le \sqrt{\frac{C\,\sigma^2_{\max}\,\mathbb{E}[k]}{N}}. \tag{5}$$

For Catalan ($\gamma = 1$) versus Chipman ($\alpha_s = 0.95, \beta = 2$):

$$\frac{\text{ECE}_{\text{Cat}}}{\text{ECE}_{\text{Chip}}} \le \sqrt{\frac{\mathbb{E}_{\text{Cat}}[k]}{\mathbb{E}_{\text{Chip}}[k]}} = \sqrt{\frac{1.373}{2.509}} = 0.740. \tag{6}$$

The Catalan prior gives a 26% tighter ECE bound at every N.

Proof. (4): By definition of ECE (calibration curve integral), $\mathrm{ECE}(k) \leq \sqrt{\sum_j \sum_c \mathbb{E}[\|\hat{p}_{jc} - p_{jc}\|^2]/k}$. Applying Proposition B.3 to each leaf with $N_j = N/k$ and using $\sigma^2_{\max} \leq 1/4$ gives (4). (5): Jensen's inequality under $\sqrt{\cdot}$ and $\mathbb{E}_k[k/N] = \mathbb{E}[k]/N$. (6): Direct substitution of $\mathbb{E}_{\mathrm{Cat}}[k] = 1.373$ and $\mathbb{E}_{\mathrm{Chip}}[k] = 2.509$. $\square$

Remark B.5. The 26% ECE advantage propagates to a 26% tighter decision-cost bound (Appendix C, Theorem C.8), and to a 26% tighter PAC-Bayes calibration penalty in the main text. The chain §2.2 → §2.5 → §2.7 (expected complexity → ECE → decision cost) gives a coherent, quantitative justification for the Catalan prior in medical classification.

C Decision-Theoretic Analysis

C.1 Optimal threshold under asymmetric loss

Let $y \in \{0, 1\}$ be the binary outcome (0=healthy, 1=OA). The loss matrix has $L(\hat{y} = 1, y = 0) = C_{FP}$ (false positive) and $L(\hat{y} = 0, y = 1) = C_{FN}$ (false negative), with $C_{FN} > C_{FP}$ for medical screening.

Theorem C.6 (Optimal threshold)**.** *The Bayes-optimal decision rule under the above loss is $d^*(x) = \mathbf{1}[P(y = 1|x) \geq \tau^*]$, where*

$$\tau^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{1}{1 + r}, \qquad r = C_{FN}/C_{FP}. \tag{7}$$

For OA detection with $r = 2$: $\tau^* = 1/3$. Standard argmax classification uses $\tau = 0.5$ and is suboptimal for all $r \neq 1$.

Proof. Minimise $\mathbb{E}[L \mid x, d] = P(y = 1|x)\, C_{FN}\, \mathbf{1}(d = 0) + P(y = 0|x)\, C_{FP}\, \mathbf{1}(d = 1)$ over $d \in \{0, 1\}$: prefer $d = 1$ iff $P(y = 1|x)(C_{FN} + C_{FP}) > C_{FP}$, i.e. $P(y = 1|x) \geq \tau^*$. $\square$

C.2 Cost saving from optimal threshold

Theorem C.7 (Asymptotic cost saving)**.** *For a k-leaf tree with true leaf probabilities $p_1, \ldots, p_k$, the asymptotic ($N \to \infty$) expected cost saving from τ^* vs argmax is*

$$\Delta EC = EC(\tau = 0.5) - EC(\tau^*) = \frac{1}{k} \sum_{i:\, p_i \in (\tau^*, 0.5)} \big[C_{FN} p_i - C_{FP}(1 - p_i)\big] \geq 0. \tag{8}$$

Proof. At large N, $\hat{p}_j \to p_j$. The two rules agree on leaves outside $(\tau^*, 0.5)$. For leaf i with $p_i \in (\tau^*, 0.5)$: argmax predicts 0 (incorrect), τ^* predicts 1 (correct). Cost under argmax: $C_{FN} p_i$; under τ^*: $C_{FP}(1 - p_i)$. The gain $C_{FN} p_i - C_{FP}(1 - p_i) = (C_{FP} + C_{FN})(p_i - \tau^*) > 0$ since $p_i > \tau^*$. $\square$

3-leaf OA example. With $p_{\text{true}} = [0.15, 0.40, 0.75]$, $C_{FP} = 1$, $C_{FN} = 2$, $\tau^* = 1/3$:

- $EC(\tau^*) = 0.383$ (Leaf 2 correctly classified as OA);
- $EC(\tau = 0.5) = EC_{\text{OC}+\tau^*} = 0.450$ (Leaf 2 classified as healthy);
- $\Delta EC = 0.067$ per patient (17.4% reduction in expected cost).

The over-confident (OC) model with $\hat{p}^{\text{OC}}(0.40) = \text{expit}(3\,\text{logit}(0.40)) = 0.229 < \tau^* = 1/3$ gains *zero* benefit from τ^*: miscalibration drags the Leaf-2 estimate below the threshold, eliminating the advantage entirely.

C.3 Miscalibration cost bound

Theorem C.8 (Miscalibration excess cost)**.** *For a miscalibrated model with estimates $\hat{p}$, the excess expected cost at threshold τ^* over the Bayes-optimal cost is*

$$\Delta EC_{\text{miscal}} \leq (C_{FP} + C_{FN}) \cdot \text{ECE\_bound}(N), \tag{9}$$

where $\text{ECE\_bound}(N) = \sqrt{C\,\mathbb{E}[k]\,\sigma^2_{\max}/N}$ *from Proposition B.4. For the Catalan vs Chipman comparison:*

$$\frac{\Delta EC_{\text{Cat}}}{\Delta EC_{\text{Chip}}} \leq 0.740 \quad \textit{(26\% tighter bound, constant in } N\textit{)}. \tag{10}$$

At $N = 40$ with $C_{FP} = 1$, $C_{FN} = 2$: $\Delta EC_{\text{Cat}} \leq 0.393$, $\Delta EC_{\text{Chip}} \leq 0.531$.

Proof. Excess cost arises when the decision based on $\hat{p}$ disagrees with the Bayes-optimal decision based on p_{true} at τ^*. For leaf i where $\hat{p}_i$ and p_i straddle τ^*, the per-leaf excess cost is at most $(C_{FP} + C_{FN})\,|p_i - \tau^*| \leq (C_{FP} + C_{FN})\,|\hat{p}_i - p_i|$. Averaging and applying $\mathbb{E}[|\hat{p} - p|] \leq \sqrt{\mathbb{E}[(\hat{p} - p)^2]} \leq \text{ECE\_bound}$ gives (9). □

Double-penalty theorem. Discriminative classifiers (e.g. random forests) incur two independent penalties: (a) wrong threshold ($\Delta EC = 0.067$ per patient for OA at $r = 2$); (b) miscalibration (≤ 0.393 per patient at $N = 40$ under Catalan). Total upper bound: $0.067 + 0.393 = 0.460$, versus the optimal $EC = 0.383$. DM-BDT at τ^* avoids both penalties simultaneously.

References

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